

K1408 — two consolidating findings from Elie, both correcting me. ★ GRAVITY IS CIRCULAR (two-way verified): every current path from the substrate scale to M_{Planck} runs through ℓ_B , which was inverted from observed G — so $G = \kappa \cdot \ell_B^2 / \pi^{n_C}$ returns $G = 1.0000000000 \cdot \ell_{\text{Planck}}^2$ (tautology to 10 decimals), and the KK reduction hands back $V_6 / \ell_B^6 = \pi^5 / n_C$, the *definition* of ℓ_B and nothing more. [?] No rearrangement produces G as an output. Cal’s §426 G -free tick is not one gate among several — on the gravity axis it IS the entire remaining work. Take gravity-forward off the route except the tick; G stays a consistency relation at 0.07%. ★ 137’s EVIDENCE is TWO COMPUTED FACTS, not a fingerprint cloud: Elie built all ~114 integer-atoms from the five primaries and 137 admits 95 representations vs neighborhood median 102 — BELOW median, 30th of 36. So the arithmetic “fingerprints” (the Fermat form, “ $137=11^2+\text{store}-16$ ”) carry zero evidential weight — I must trim them. The only real evidence is the quadric $27=N_c^3$ (unique at $n_C=5$) + the K38 +rank — both computed geometry. Tier unchanged (Identified-strong); the argument gets honest. Cal cold-reads.

Keeper (2026-08-12, ~13:30. Elie proved gravity’s chain circular and quantified that my fingerprint argument was mostly noise. Both are self-corrections I owe — and both leave the picture cleaner and more defensible. Cal cold-reads. Nothing pushed.)

★ Finding 1: the gravity chain is CIRCULAR — G -forward reduces to the tick ONLY (Elie, verified two ways)

Elie closed the substrate-scale $\rightarrow M_{\text{Planck}} \rightarrow G$ line and it’s circular by construction: - The KK route: put the T1301 reduction in, solve for the compactification volume $\rightarrow V_6 / \ell_B^6 = \pi^5 / n_C$ exactly — the relation hands back the *definition* of ℓ_B and nothing else, because ℓ_B was inverted from observed G in the first place. - The direct route: the corpus’s own $G = \kappa \cdot \ell_B^2 / \pi^{n_C}$ returns $G = 1.0000000000 \cdot \ell_{\text{Planck}}^2$ — a tautology to ten decimals. - [?] Every current path from the substrate scale to Newton’s G runs through a number obtained by working backwards from G . No rearrangement of those paths will ever produce G as an output. This is the concrete, two-way confirmation of my own K1374/K1394 circularity audit — now proven, not suspected. - ★ The consequence (the useful part): Cal’s §426 G -free tick is not “one gate among several” — on the gravity axis it IS the entire remaining work. Everything else comes out however the tick comes out. So: G is a CONSISTENCY RELATION at 0.07% (real, tight — not nothing), and it becomes a forward *prediction* if and only if the G -free tick is derived from the geometry with a non-gravitational pin (K1394). That is hard (maybe not achievable); if it can’t be, G stays a consistency relation — which is honest and still a tight result. Take gravity-forward off the active route except the single tick computation. (Elie closed the line; no follow-on toys; the next step is a spectral computation belonging to whoever forces the tick. Route item 3 comes off Elie.)

★ Finding 2: 137’s evidence is TWO computed facts — the fingerprint cloud is noise (Elie; I self-correct)

I’ve been citing “137 leaves fingerprints everywhere $\rightarrow$ it’s an invariant, not a fit” (K1401/K1405/K1406). Elie quantified it, and the arithmetic half is worthless: - He built every atom the five integers make (~114) and counted sums of ≤ 3 : 137 admits 95 representations against a neighborhood median of 102 — it is BELOW median, 30th of 36, robust across five grammar variants. So “ $137 = N_c^3 \cdot n_C + \text{rank}$ ” carries zero evidential weight — every

integer near there has ~100 such forms, and 137 has slightly *fewer* than its neighbors. This is “a prime has no forced decomposition” *with a number*. - The Fermat item is mild (6.1%, one prime in 16 shares it), and “ $137 = 11^2 + \text{store-16}$ ” is the SAME arithmetic fact as the Fermat small part ($4^2 = 16 = \text{rank}^4$) — one fact, not two (count-once, applied to Elie’s own morning result — and to my K1401/K1405 banking of it). Drop “ $137 = 11^2 + \text{store-16}$ ” as independent corroboration; it’s the Fermat form. - ★ Exactly TWO fingerprints survive, and they’re the two that were COMPUTED, not noticed: (1) the quadric’s $h^0(Q^5, O(2)) = 27 = N_c^3$, unique at $n_c=5$ (Hilbert polynomial, target-innocent — Cal’s table 14/20/27/35); (2) the K38 +rank (three convergent geometric routes). Both are geometry, both target-innocent — and they sandwich the open $\times 5$ (which is why Grace’s refutation was informative, not a failure). - ¶ SELF-CORRECTION to my framing: the basis for “137 is a computed invariant of D_{IV}^5 ” is TWO computed geometric facts, NOT a cloud of arithmetic fingerprints. The tier is UNCHANGED (Identified-strong — two target-innocent computed facts is real evidence), but the argument must be trimmed to those two facts. “A conclusion propped by one good pillar and three rotten ones gets judged on the rotten ones” (Elie) — so present the two computed pillars, drop the arithmetic scaffolding. *A referee runs the representation-count over coffee; we run it first.* - ★ The referee sentence to put in the text (Elie’s, ratified): “ $n_c = 5$ is in the name of the domain — where is 137 in its definition?” Honest answer: $n_c = 5$ is an input (a primitive label); 137 is a claimed *computed* invariant whose computation our own audit says is OPEN (Cal #27, K1407). The tier already carries exactly that — nothing changes but the wording. State it straight; it survives the question.

Ratify Cal’s parity verdict (§435) — principled, with a standing guard

Elie’s “amplitude-ratios even / loop-orders odd” repair PASSES Cal’s hostile read: principled, not author-defended — non-circular (the Lamb shift is loop-order *by its physics*, independent of noticing α^5 is odd), textbook mechanism (Born rule = $|\text{amplitude}|^2 = \text{even}$; loop corrections = vertex-count = parity-free). Cost banked honestly: the strong falsifier (“any observable at odd $\alpha^{(-137)}$ kills it”) is DEAD; the survivor is narrower — “any *Born-rule-type* observable at an odd rung kills it, loop corrections exempt”; $\alpha^{(-137)/\alpha}(-274)$ still tests ($274 = 2 \cdot 137$, even). ★ Standing guard (Cal, ratified): it turns author-defended the instant a future odd-rung counterexample is reclassified as “loop-order” to save the rule — classify every quantity’s bucket by DERIVATION, committed before examining parity, never parity-first. Cal holds it on the next candidate.

Ratify Lyra’s B1 collapse (§ to one dictionary check) + Cal’s gates

Cal’s “prove same *operator*, not same *kind*” demand collapsed B1’s same-object leg to ONE dictionary check: does Finster’s “occupied” mean our half-turn (spinor) modes? Two projectors are equal iff they keep the same states; if the ranges match, they’re the same operator and his causal rule falls out for free (same filter). Cal’s constructed-artifact gates still owed on the built map (§433): $P^2=P$ and $P\ddagger=P$ in the Krein metric, A_{xy} signature = Finster’s causal eigenvalues, equality-not-category — and the minimum is the unrun mountain (our config is the bottom of his action, genuinely open; the note Lyra leaned on is a different functional). B1 = one dictionary check away from same-object; the minimum is the real climb. Gates clear on the map, not the intention.

The honest foundation (final, consolidated — state it this way)

ONE input (the geometry is D_{IV}^5) $\rightarrow -\alpha = 1/N_{\text{max}}$: Identified-strong — a non-tunable computed invariant, evidenced by TWO target-innocent computed facts (quadric $27=N_c^3$ unique@ $n_c=5$; K38 +rank), mechanism-open (the $\times 5$ / Gaussian-norm simmer, one hand). - m_e exponent $2C_2 = 12$: convention-pinned (standard $1/16\pi G$, 6 corpus locations). - G: consistency relation at 0.07% (circular by construction, proven two ways) — forward iff the G-free tick (Cal §426, non-grav pin K1394) is derived; that single computation is the entire gravity-forward work. - The spearhead: B1/C4 (one dictionary check + the minimum) + the QM papers.

Route

1. ★ Lyra + Cal (+ Elie) — B1/C4: the ONE dictionary check (Finster’s “occupied” = our half-turn modes?); Elie builds the $P^2=P/P\ddagger=P$ Krein check on the $g=7$ fermionic projector (§433) as a toy while Lyra/Cal do the map; then the minimum.
2. Whoever’s free — the QM papers (writable now, 137-free).

3. The G-free tick — the *entire* gravity-forward work (spectral, non-grav pin); belongs to whoever forces it; not Elie’s follow-on. Until then G is an honest consistency relation.
4. Grace — the Gaussian-norm simmer (one hand) + B1/QM graph support; trim the data-layer 137 evidence to the two computed facts.
5. Keeper — self-corrected the fingerprint framing (two computed facts, drop the arithmetic cloud + “137=11²+store-16” duplicate); ratified gravity-is-circular (G-forward = tick only) + the parity guard + the B1 collapse; recused.

— Keeper, K1408, 2026-08-12. Elie 2 findings, both correct me. (1) GRAVITY CIRCULAR (2-way verified): KK reduction $\rightarrow V_6/\ell_B = \pi^5/n_C$ (hands back ℓ_B ’s definition); $G = \kappa \ell_B^2 / \pi^{n_C} = 1.000000 \cdot \ell_{\text{Planck}}^2$ (tautology 10 decimals); ℓ_B inverted from observed $G \rightarrow$ NO path produces G as output $\rightarrow$ Cal §426 G-free tick IS the entire gravity-forward work (not one gate); G = consistency relation 0.07% (real), forward iff G-free tick w/ non-grav pin (K1394, hard); take gravity-forward off route except the tick; validates K1374/K1394. (2) 137 EVIDENCE = 2 COMPUTED FACTS not a cloud: Elie built ~114 integer-atoms, 137 = 95 reps vs median 102 = BELOW median (30/36) $\rightarrow$ arithmetic fingerprints ZERO weight; Fermat mild (6.1%); “137=11²+store-16” = SAME fact as Fermat (count-once) — DROP as independent. Survivors: quadric $27 = N_c^3$ unique@ $n_C = 5$ + K38 +rank (both computed geometry, sandwich the open $\times 5$). SELF-CORRECT my K1401/05/06 fingerprint framing $\rightarrow$ trim to 2 computed pillars; tier UNCHANGED (Identified-strong). Referee sentence: “ $n_C = 5$ in the name; where’s 137 in the definition?” $\rightarrow$ 5 input, 137 computed-invariant-mechanism-open. RATIFY Cal parity verdict (principled non-circular; strong falsifier dead, survives Born-rule-type-at-odd-rung; guard: bucket-by-derivation-before-parity). RATIFY Lyra B1 collapse: 1 dictionary check (Finster “occupied”=our half-turn modes?)=same operator; Cal gates $P^2 = P/P \ddagger = P$ Krein + A_{xy} signature + equality-not-category; minimum = unrun mountain. Route: Lyra+Cal+Elie B1 (dictionary check + Elie Krein $P^2 = P$ toy); QM papers; G-free tick = entire gravity-forward work; Grace Gaussian simmer + trim data 137 evidence; Keeper self-corrected+recused. Cal cold-reads. Nothing pushed.